

Rakuten MLOps Project – Current Status

Hi everyone,

I set up the initial project repository and completed **Step 1 (Project Setup)** and **Step 2 (Data Inspection & Basic Preprocessing)** so that we all start from the same clean structure.

Repository:

<https://github.com/sosu0307/rakuten-product-classification-mlops>

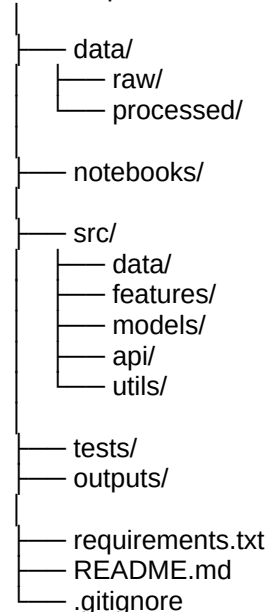
Please **clone the repository** and follow the instructions below so that we all work with the same setup.

Step 1 – Project Setup (Completed)

The repository already contains:

Project structure

rakuten-product-classification-mlops



Also included:

- Python environment structure
- initial requirements.txt
- .gitignore
- base scripts for loading and inspecting the data

Important – Data Handling

The dataset **must not be committed to GitHub**.

Each team member should store the dataset **locally only**.

After cloning the repo, place the files there: data/raw/

Rename the dataset files

The Rakuten files include random suffixes.

Please rename them exactly like this so that all scripts work:

```
X_train_XXXX.csv → X_train.csv  
X_test_XXXX.csv → X_test.csv  
y_train_XXXX.csv → y_train.csv  
y_random_XXXX.csv → y_random.csv
```

Place them in: data/raw/

Images should also be placed in the same folder.

Step 2 – Data Inspection & Basic Preprocessing (**Completed**)

The following checks and preprocessing steps were implemented:

Dataset sizes

Train samples	Test samples
212,120	37,347

Input columns

image_file_name
item_name
item_caption

Target column

color_tags

Missing values

item_caption has missing values (~22k rows)

This was handled by filling empty strings.

Text preparation

A combined text field was created:

```
combined_text = item_name + item_caption
```

This will be used for the baseline text model.

Label structure

color_tags was converted from string format to a list.

Problem type

This is a **multi-label classification problem**
(one product can have multiple colors).

Number of color classes

19 unique color labels:

Beige
Black
Blue
Brown
Burgundy
Gold
Green
Grey
Khaki
Multiple Colors
Navy
Orange
Pink
Purple
Red
Silver
Transparent
White
Yellow

Most frequent labels

Top colors in the dataset:

Black
White
Blue
Brown
Grey
Beige
Navy
Pink
Red
Green

Important Reminder

The mentor explicitly mentioned that we **should not spend too much time on model tuning or preprocessing**.

The goal is mainly to build a **working end-to-end pipeline**, not a perfect model.

So we should aim to:

1. get a working baseline quickly
2. then move to the MLOps architecture

Feel free to start with **Step 3 or Step 4**.

Let me know if something in the setup doesn't work after cloning the repo.